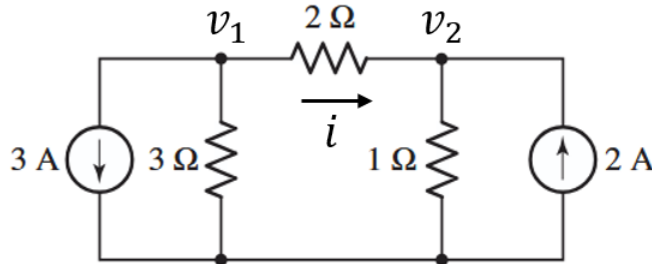


HW 3 - Zeshui Song

4.6 6. Calculate the power dissipated in the $1\ \Omega$ resistor of Fig. 4.36.



■ **FIGURE 4.36** *Ref.*

Using nodal analysis at node 1:

$$[-3\text{ A}] = \frac{v_1}{[3\ \Omega]} + \frac{v_1 - v_2}{[2\ \Omega]} \quad \Rightarrow \quad -3 = \frac{5}{6}v_1 - \frac{1}{2}v_2$$

At node 2:

$$\frac{v_1 - v_2}{[2\ \Omega]} = \frac{v_2}{[1\ \Omega]} - [2\text{ A}] \quad \Rightarrow \quad -2 = \frac{1}{2}v_1 - \frac{3}{2}v_2$$

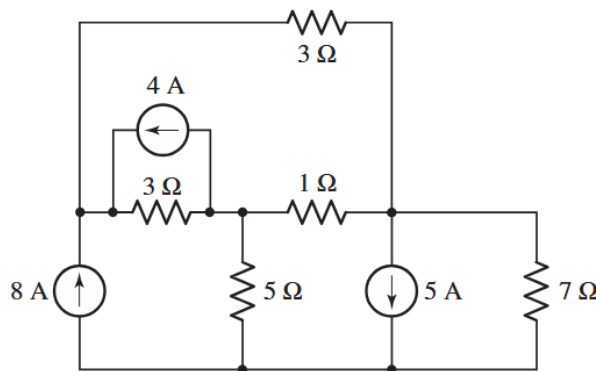
Solving the system of equations, we get $v_1 = -7/2\text{ V}$ and $v_2 = 1/6\text{ V}$.

The power dissipated by the $1\ \Omega$ resistor is then

$$P = VI = \frac{v_2^2}{R} = \frac{[1/6\text{ V}]^2}{[1\ \Omega]} = 0.02777\text{ W}$$

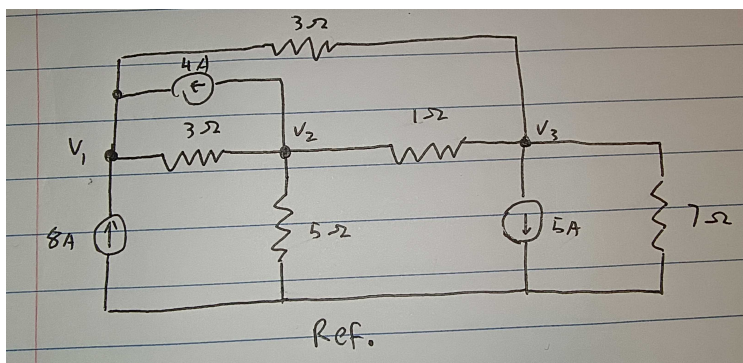
$P = 0.0278\text{ W}$

4.12 12. Using the bottom node as reference, determine the voltage across the $5\ \Omega$ resistor in the circuit of Fig. 4.42, and calculate the power dissipated by the $7\ \Omega$ resistor.



■ **FIGURE 4.42**

Redrawing the circuit slightly and labeling the nodes:



At node 1:

$$[8 \text{ A}] = \frac{v_1 - v_3}{[3 \Omega]} - [4 \text{ A}] + \frac{v_1 - v_2}{[3 \Omega]} \quad \Rightarrow \quad 12 = \frac{2}{3}v_1 - \frac{1}{3}v_2 - \frac{1}{3}v_3$$

At node 2:

$$\frac{v_1 - v_2}{[3 \Omega]} = [4 \text{ A}] + \frac{v_2 - v_3}{[1 \Omega]} + \frac{v_2}{[5 \Omega]} \quad \Rightarrow \quad 4 = \frac{1}{3}v_1 - \frac{23}{15}v_2 + v_3$$

At node 3:

$$\frac{v_2 - v_3}{[1 \Omega]} + \frac{v_1 - v_3}{[3 \Omega]} = [5 \text{ A}] + \frac{v_3}{[7 \Omega]} \quad \Rightarrow \quad 5 = \frac{1}{3}v_1 + v_2 - \frac{31}{21}v_3$$

Solving the system of equations, we get $v_1 = 26.733 \text{ V}$, $v_2 = 8.833 \text{ V}$, and $v_3 = 8.633 \text{ V}$.

The power dissipated by the 7Ω resistor is then

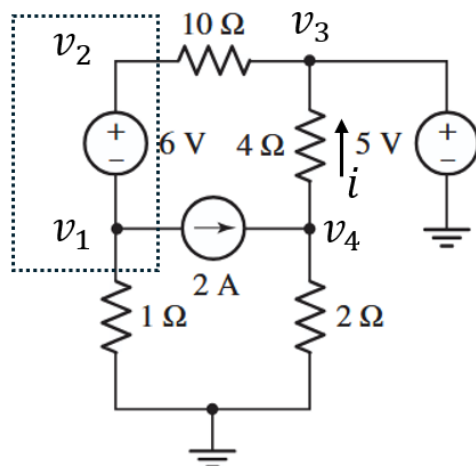
$$P = VI = \frac{v_3^2}{R} = \frac{[8.633 \text{ V}]^2}{[7 \Omega]} = 10.54 \text{ W}$$

$$\boxed{P = 10.64 \text{ W}}$$

and the voltage across the 5Ω resistor is

$$\boxed{v_3 = 8.633 \text{ V}}$$

4.20 20. For the circuit of Fig. 4.50, determine all four nodal voltages.



■ **FIGURE 4.50**

Using a super node around the voltage source at nodes 1 and 2:

$$v_2 - v_1 = [6 \text{ V}] \quad \Rightarrow \quad 6 = -1v_1 + v_2$$

$$\frac{0 - v_1}{[1 \Omega]} = \frac{v_2 - v_3}{[10 \Omega]} + [2 \text{ A}] \quad \Rightarrow \quad 2 = -v_1 - \frac{1}{10}v_2 + \frac{1}{10}v_3$$

At node 3:

$$v_3 = [5 \text{ V}]$$

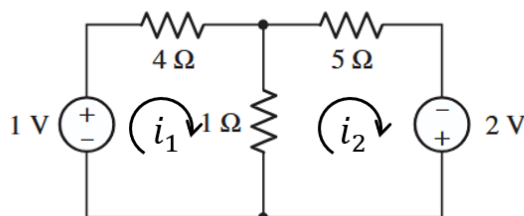
At node 4:

$$[2 \text{ A}] = \frac{v_4}{[2 \Omega]} + \frac{v_4 - v_3}{[4 \Omega]} \quad \Rightarrow \quad 2 = -\frac{1}{4}v_3 + \frac{3}{4}v_4$$

Solving the system of equations, we get

$$v_1 = -1.909 \text{ V}, v_2 = 4.0909 \text{ V}, v_3 = 5 \text{ V}, \text{ and } v_4 = 4.333 \text{ V}$$

4.32 32. Determine the currents flowing out of the positive terminal of each voltage source in the circuit of Fig. 4.60.



■ **FIGURE 4.60**

Using mesh analysis on the left loop:

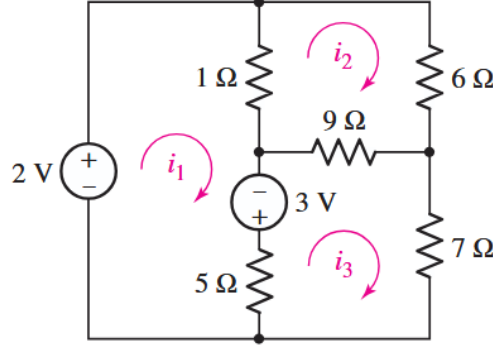
$$[1 \text{ V}] - [4 \Omega]i_1 - [1 \Omega](i_1 - i_2) = 0 \quad \Rightarrow \quad 1 = 5i_1 - i_2$$

Using mesh analysis on the right loop:

$$-[1\ \Omega](i_2 - i_1) - [5\ \Omega]i_2 + [2\ \text{V}] = 0 \quad \Rightarrow \quad 2 = -i_1 + 6i_2$$

Solving the system of equations, we get $i_1 = 0.2758\ \text{A}$ and $i_2 = 0.3793\ \text{A}$.

4.36 36. Calculate the power dissipated by each resistor in the circuit of Fig. 4.63.



■ FIGURE 4.63

Mesh 1:

$$[2\ \text{V}] - [1\ \Omega](i_1 - i_2) + [3\ \text{V}] - [5\ \Omega](i_1 - i_3) = 0 \quad \Rightarrow \quad 5 = 6i_1 - i_2 - 5i_3$$

Mesh 2:

$$[1\ \Omega](i_2 - i_1) - [6\ \Omega]i_2 - [9\ \Omega](i_2 - i_3) \quad \Rightarrow \quad 0 = i_1 + 14i_2 - 9i_3$$

Mesh 3:

$$-[5\ \Omega](i_3 - i_1) - [3\ \text{V}] - [9\ \Omega](i_3 - i_2) - [7\ \Omega]i_3 = 0 \quad \Rightarrow \quad -3 = -5i_1 - 9i_2 + 21i_3$$

Solving the system of equations, we get $i_1 = 0.8724\ \text{A}$, $i_2 = -0.0284\ \text{A}$, and $i_3 = 0.0526\ \text{A}$.

Power dissipated by the $1\ \Omega$ resistor:

$$P_{1\Omega} = Vi = (i_1 - i_2)^2 R = [0.8724 - (-0.0284)\ \text{A}]^2 \cdot [1\ \Omega] = 0.8114\ \text{W}$$

Power dissipated by the $5\ \Omega$ resistor:

$$P_{5\Omega} = Vi = (i_1 - i_3)^2 R = [0.8724 - 0.0526\ \text{A}]^2 \cdot [5\ \Omega] = 3.360\ \text{W}$$

Power dissipated by the $9\ \Omega$ resistor:

$$P_{9\Omega} = Vi = (i_3 - i_2)^2 R = [0.0526 - (-0.0284)\ \text{A}]^2 \cdot [9\ \Omega] = 0.0590\ \text{W}$$

Power dissipated by the $6\ \Omega$ resistor:

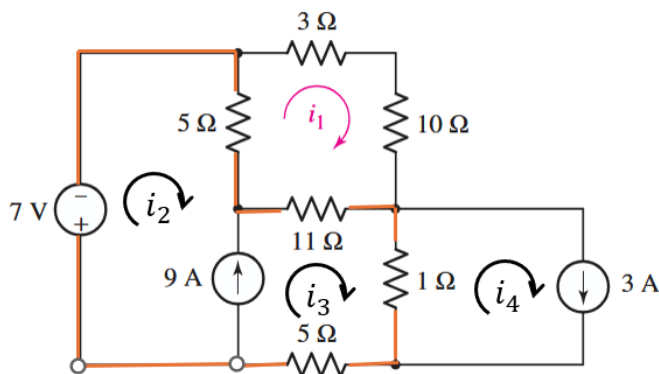
$$P_{6\Omega} = Vi = (i_2)^2 R = [0.0284\ \text{A}]^2 \cdot [6\ \Omega] = 0.0048\ \text{W}$$

Power dissipated by the $7\ \Omega$ resistor:

$$P_{7\Omega} = Vi = (i_3)^2 R = [0.0526\ \text{A}]^2 \cdot [7\ \Omega] = 0.0193\ \text{W}$$

4.46

46. For the circuit of Fig. 4.73, determine the mesh current i_1 and the power dissipated by the $1\ \Omega$ resistor.



■ FIGURE 4.73

Orange supermesh:

$$i_3 - i_2 = [9\text{ A}] \implies 9 = -i_2 + i_3$$

$$- [7\text{ V}] - [5\ \Omega](i_2 - i_1) - [11\ \Omega](i_3 - i_1) - [1\ \Omega](i_3 - i_4) - [5\ \Omega]i_3 = 0 \implies 7 = 16i_1 - 5i_2 - 17i_3 + i_4$$

Mesh 1:

$$- [5\ \Omega](i_1 - i_2) - [3\ \Omega]i_1 - [10\ \Omega]i_1 - [11\ \Omega](i_1 - i_3) = 0 \implies -29i_2 + 5i_2 + 11i_3 = 0$$

Mesh 4:

$$i_4 = [3\text{ A}] \implies 3 = i_4$$

Solving the system of equations, we get $i_1 = -0.8743\text{ A}$, $i_2 = -7.7722\text{ A}$, $i_3 = 1.2277\text{ A}$, and $i_4 = 3\text{ A}$.

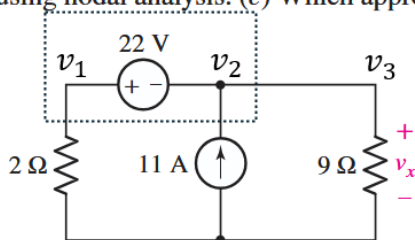
Power dissipated by the $1\ \Omega$ resistor:

$$P_{1\Omega} = Vi = (i_3 - i_4)^2 R = [1.2277 - 3\text{ A}]^2 \cdot [1\ \Omega] = 3.141\text{ W}$$

$$i_1 = -0.8743\text{ A}$$

4.56

56. Solve for the voltage v_x as labeled in the circuit of Fig. 4.82 using (a) mesh analysis. (b) Repeat using nodal analysis. (c) Which approach was easier, and why?



■ FIGURE 4.82 Ref.

(a) Mesh analysis

Supermesh:

$$-[2\Omega]i_1 - [22\text{ V}] - [9\Omega](i_2) = 0 \quad \Rightarrow \quad 22 = -2i_1 - 9i_2$$

$$i_2 - i_1 = [11\text{ A}] \quad \Rightarrow \quad 11 = -i_1 + i_2$$

Solving the system of equations, we get $i_1 = -11\text{ A}$ and $i_2 = 0\text{ A}$.

Thus the voltage across the 9Ω resistor is

$$v_{9\Omega} = i_2 \cdot R = 0\text{ A} \cdot [9\Omega] = 0\text{ V}$$

(b) Node analysis

At the 1-2 supernode:

$$v_1 - v_2 = [22\text{ V}] \quad \Rightarrow \quad 22 = v_1 - v_2$$

$$[11\text{ A}] = \frac{v_1}{[2\Omega]} + \frac{v_2}{[9\Omega]} \quad \Rightarrow \quad 11 = \frac{1}{2}v_1 + \frac{1}{9}v_2$$

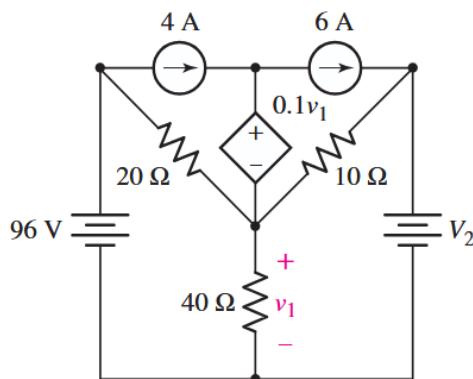
Node 3 is connected to the supernode, so there is no KCL equation for it. Solving the system of equations, we get $v_1 = 22\text{ V}$, $v_2 = 0\text{ V}$, and $v_3 = v_2 = 0\text{ V}$.

Thus the voltage across the 9Ω resistor is

$$v_{9\Omega} = v_3 = 0\text{ V}$$

(c) The mesh analysis was easier for me because it's the same amount of equations, but the supernode in the node analysis got me tripped up.

- 4.57** 57. Consider the five-source circuit of Fig. 4.83. Determine the total number of simultaneous equations that must be solved in order to determine v_1 using (a) nodal analysis; (b) mesh analysis. (c) Which method is preferred, and does it depend on which side of the 40Ω resistor is chosen as the reference node? Explain your answer.



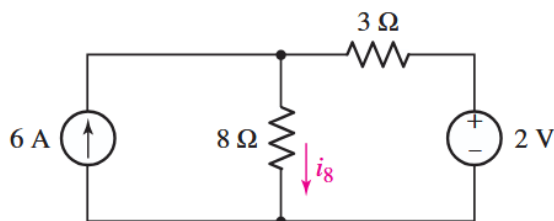
■ FIGURE 4.83

(a) For nodal analysis, we have a supernode around the $0.1v_1$ voltage source and if we place reference node bottom wire, then we will have defined the leftmost and rightmost nodes. Thus we will only have 2 eqns to solve for with the supernode.

(b) For mesh analysis, we would have a supermesh after cutting out the 2 current sources. This would leave us with 2 equations to solve for the supermesh as the current in from the current sources are known.

(c) I think the nodal analysis is easier for this problem because the multiple current sources makes the mesh analysis more difficult. The reference node needs to be on the negative terminal of the $40\ \Omega$ resistor because it is needed to remove 2 unknown voltages.

5.3 3. Considering the circuit of Fig. 5.48, employ superposition to determine the two components of i_8 arising from the action of the two independent sources, respectively.



■ **FIGURE 5.48**

Turning off the voltage source (use node analysis):

$$[6\text{ A}] = \frac{v_1}{[8\ \Omega]} + \frac{v_1}{[3\ \Omega]} \quad \Rightarrow \quad v_1 = [13.090\text{ V}]$$

$$i'_8 = \frac{v_1}{[8\ \Omega]} = \frac{13.090}{8} = [1.6363\text{ A}]$$

Turning off the current source:

$$R_{\text{eq}} = [3\ \Omega] + [8\ \Omega] = [11\ \Omega]$$

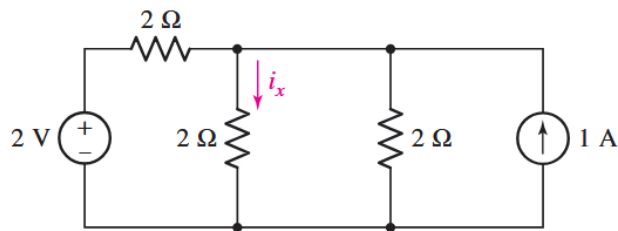
$$i''_8 = \frac{[2\text{ V}]}{[11\ \Omega]} = [0.1818\text{ A}]$$

Superposition:

$$i_8 = i'_8 + i''_8 = [1.6363 + 0.1818]\text{ A} = [1.8181\text{ A}]$$

5.5

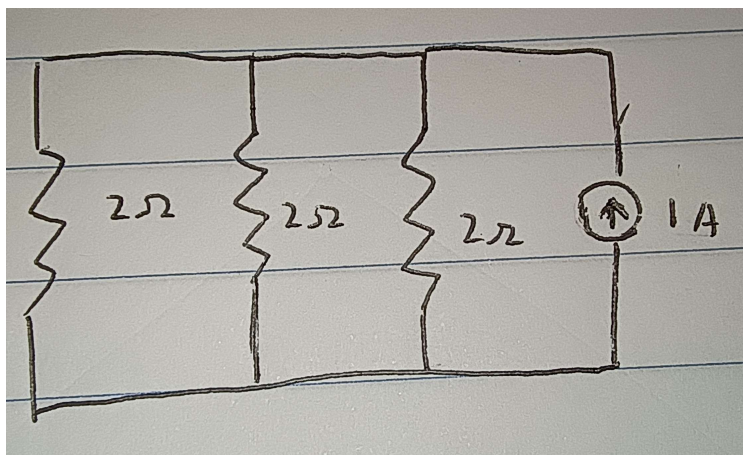
5. (a) Using superposition to consider each source one at a time, compute i_x .
 (b) Determine the percentage of i_x arising from each source. (c) Adjusting only the current source, alter the circuit to double i_x .



■ FIGURE 5.50

(a)

Turning off the voltage source:



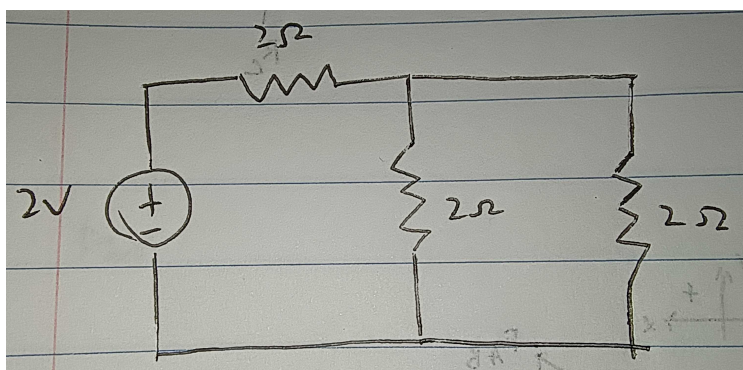
Combine the other two parallel resistors:

$$R_{eq} = \frac{[2 + 2\Omega]}{[2 \cdot 2\Omega]} = 1\Omega$$

Using the current division rule:

$$i'_x = \frac{[1\Omega]}{[1 + 2\Omega]} \cdot [1\text{ A}] = \frac{1}{3} \cdot 1 = 0.3333\text{ A}$$

Turning off the current source:



Combine the parallel resistors:

$$R_{eq} = \frac{[2 + 2\ \Omega]}{[2 \cdot 2\ \Omega]} = 1\ \Omega$$

Use voltage division to find voltage across the $1\ \Omega$ resistor:

$$v_1 = [2\ \text{V}] \frac{[1\ \Omega]}{[1 + 2\ \Omega]} = 0.6666\ \text{V}$$

This is the same voltage across the original $2\ \Omega$ resistor, so we can find the current through it:

$$i''_x = \frac{v_1}{[2\ \Omega]} = \frac{0.6666}{2} = 0.3333\ \text{A}$$

Superposition:

$$i_x = i'_x + i''_x = [0.3333 + 0.3333\ \text{A}] = 0.6666\ \text{A}$$

(b)

The current contributions are equal so its 50% from the voltage source and 50% from the current source. (c) To double the current, the contribution from the current source will have to be $i'_x = 1\ \text{A}$.

$$i'_x = [1\ \text{A}] = \frac{[1\ \Omega]}{[1 + 2\ \Omega]} \cdot i_s \implies i_s = 3\ \text{A}$$

5.11 11. Employ superposition principles to obtain a value for the current I_x as labeled in Fig. 5.56.

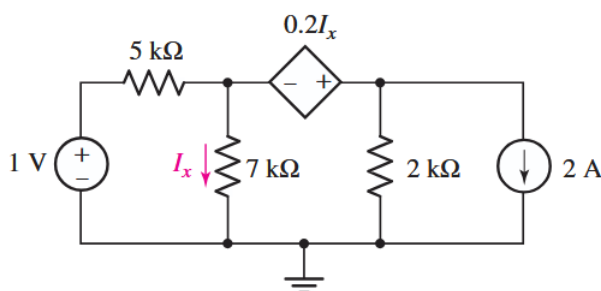
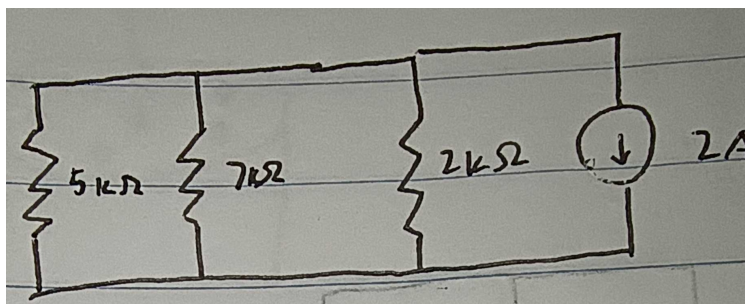


FIGURE 5.56

Isolating current source:



Combining the parallel resistors:

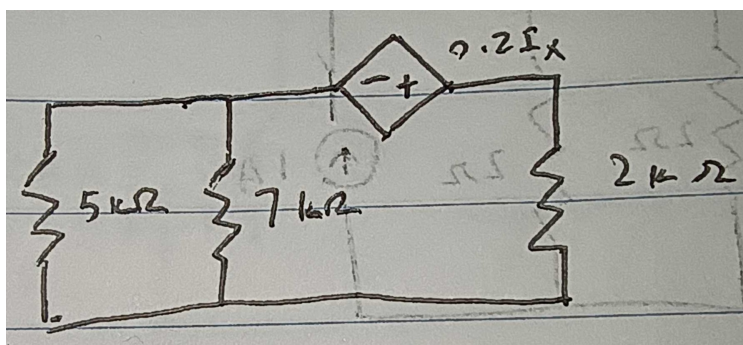
$$\frac{1}{R_{eq}} = \frac{1}{[2\text{ k}\Omega]} + \frac{1}{[7\text{ k}\Omega]} = \frac{1}{[5\text{ k}\Omega]} \quad \Rightarrow \quad R_{eq} = 1.1864\text{ k}\Omega$$

$$\Rightarrow v = [2\text{ A}] \cdot [1.1864\text{ k}\Omega] = 2.3728\text{ V}$$

This is equal to the voltage over the $7\text{ k}\Omega$ resistor, so we can find the current through it:

$$I'_x = \frac{v}{[7\text{ k}\Omega]} = \frac{2.3728}{7000} = 3.389 \times 10^{-4}\text{ A} \uparrow$$

Isolating dependent voltage source:



Combining the parallel resistors:

$$R_{eq} = \frac{[5 \cdot 7\text{ k}\Omega]}{[5 + 7\text{ k}\Omega]} = 2.9166\text{ k}\Omega$$

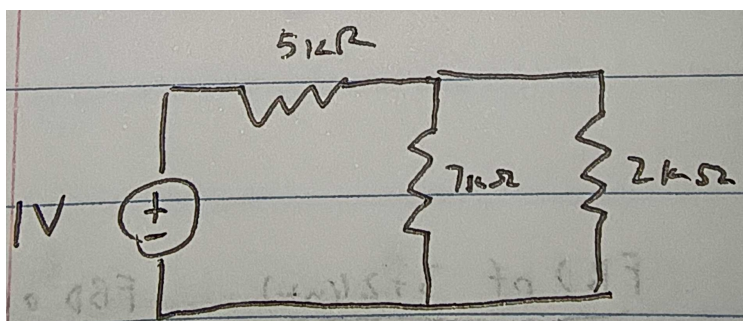
Current voltage division:

$$v = \frac{[2916.6\text{ }\Omega]}{[2916.6\text{ }\Omega + 2000\text{ }\Omega]} \cdot [0.2I_x\text{ V}] = 0.1186I_x\text{ V}$$

This is equal to the voltage across the $7\text{ k}\Omega$ resistor, so we can find the current through it:

$$I''_x = \frac{v}{[7\text{ k}\Omega]} = \frac{0.1186I_x}{7000} = 1.6949 \times 10^{-5}I_x\text{ A} \uparrow$$

Isolating independent voltage source:



Combining the parallel resistors:

$$R_{eq} = \frac{[2 \cdot 7 \text{ k}\Omega]}{[2 + 7 \text{ k}\Omega]} = 1.5555 \text{ k}\Omega$$

Current voltage division:

$$v = \frac{[1555.55 \Omega]}{[1555.55 \Omega + 5000 \Omega]} \cdot [1 \text{ V}] = 0.2372 \text{ V}$$

This is equal to the voltage across the 7 kΩ resistor, so we can find the current through it:

$$I_x''' = \frac{v}{[7 \text{ k}\Omega]} = \frac{0.2372}{7000} = 3.3898 \times 10^{-5} \text{ A} \downarrow$$

Superposition:

$$I_x = I_x' + I_x'' + I_x''' = [3.389 \times 10^{-4} + 1.6949 \times 10^{-5} I_x - 3.3898 \times 10^{-5} \text{ A}]$$

$$\Rightarrow 0.9999 I_x = 3.2262 \times 10^{-4} \text{ A} \quad \Rightarrow \quad \boxed{I_x = 3.0500 \times 10^{-4} \text{ A} \uparrow}$$

- 5.18** 18. (a) Using repeated source transformations, reduce the circuit of Fig. 5.62 to a voltage source in series with a resistor, both of which are in series with the 6 MΩ resistor. (b) Calculate the power dissipated by the 6 MΩ resistor using your simplified circuit.

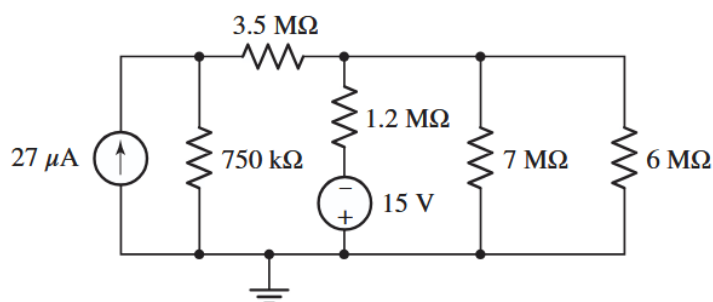
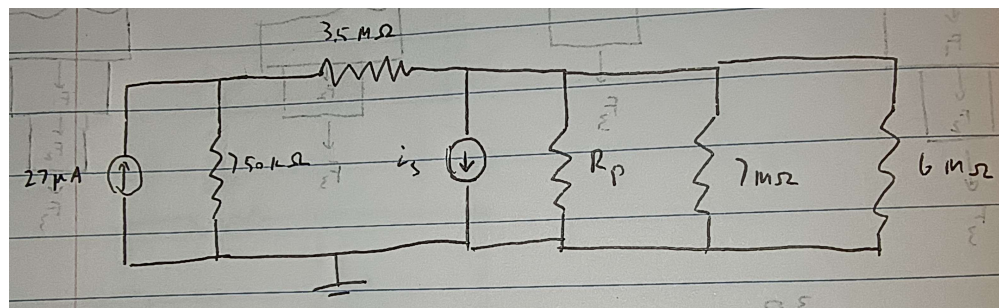


FIGURE 5.62

(a) Goal: put everything in parallel to each other with 2 current sources and 2 resistors (one of them being 6 MΩ).

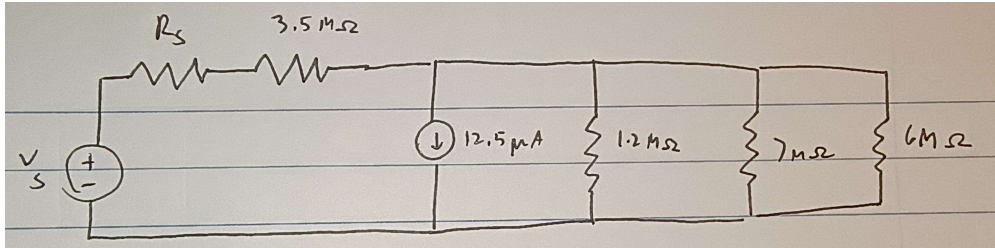
Starting by transforming the voltage source into a current source:



$$R_s = R_p \implies R_p = 1.2 \text{ M}\Omega$$

$$v_s = i_s R_s \implies i_s = \frac{[15 \text{ V}]}{[1.2 \text{ M}\Omega]} = 12.5 \times 10^{-6} \text{ A}$$

Converting the $27 \mu\text{A}$ current source to a voltage source:



$$R_s = R_p \implies R_s = 750 \text{ k}\Omega$$

$$v_s = i_s R_s \implies v_s = [27 \times 10^{-6} \text{ A}] \cdot [750 \times 10^3 \Omega] = 20.25 \text{ V}$$

Now we can combine the two resistors in series:

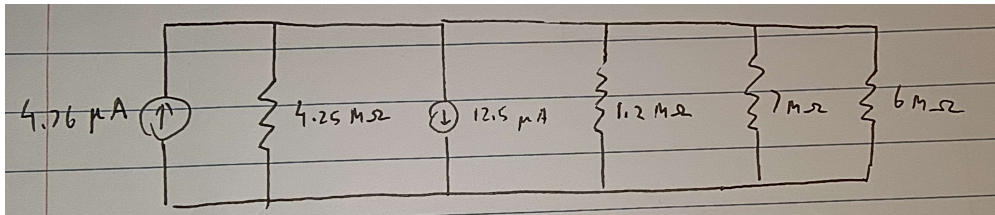
$$R = [750 \times 10^3 \Omega] + [3.5 \times 10^6 \Omega] = 4.25 \times 10^6 \Omega$$

At the 20.25 V voltage source is in series with the $4.25 \text{ M}\Omega$ resistor, we can convert it to a current source:

$$R_s = R_p \implies R_p = 4.25 \text{ M}\Omega$$

$$v_s = i_s R_s \implies i_s = \frac{[20.25 \text{ V}]}{[4.25 \text{ M}\Omega]} = 4.7647 \times 10^{-6} \text{ A}$$

Now everything is in parallel:



Combine the parallel current sources (taking up as positive):

$$i_{\text{eq}} = [4.7647 \times 10^{-6} \text{ A}] - [12.5 \times 10^{-6} \text{ A}] = -7.7353 \times 10^{-6} \text{ A}$$

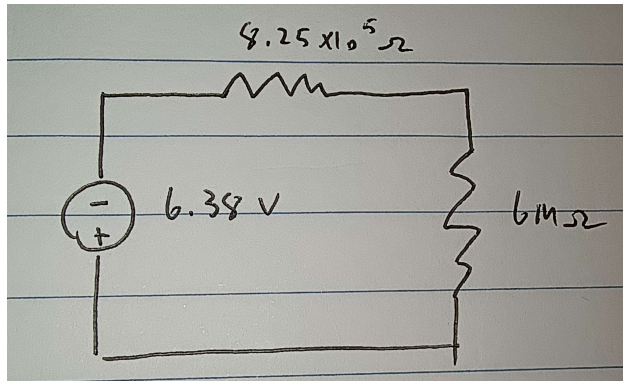
Combining the parallel resistors (except the $6 \text{ M}\Omega$ one):

$$\frac{1}{R_{\parallel}} = \frac{1}{[4.25 \times 10^6 \Omega]} + \frac{1}{[1.2 \times 10^6 \Omega]} + \frac{1}{[7 \times 10^6 \Omega]} \implies R_{\parallel} = 8.2543 \times 10^5 \Omega$$

This can finally be converted to a voltage source:

$$R_s = R_p \implies R_s = 8.2543 \times 10^5 \Omega$$

$$v_s = i_s R_s \implies v_s = [-7.7353 \times 10^{-6} \text{ A}] \cdot [8.2543 \times 10^5 \Omega] = -6.3849 \text{ V}$$



(b) Using voltage division to find voltage across the $6\text{ M}\Omega$ resistor:

$$v = \frac{[6 \times 10^6 \Omega]}{[6 \times 10^6 \Omega + 8.2543 \times 10^5 \Omega]} \cdot [6.3849 \text{ V}] = 5.6127 \text{ V}$$

Power dissipated by the $6\text{ M}\Omega$ resistor:

$$P = \frac{v^2}{R} = \frac{[5.6127 \text{ V}]^2}{[6 \times 10^6 \Omega]} = \boxed{5.25 \times 10^{-6} \text{ W}}$$